

1-A:

A) The project's **social cost is \$100 million**, which includes the **private cost of \$40 million**. This means the **external cost is \$60 million (\$100 million - \$40 million)**.

On the other hand, the project has an external benefit of only \$20 million. Since the external cost (\$60 million) is greater than the external benefit (\$20 million), we can conclude that the project's external costs outweigh its external benefits.

B) This is incorrect because the private cost of \$40 is not greater than external cost of \$60.

C) Since the net social value, i.e. $SB - SC = 0$, this represents that the social benefits will equal to social costs which are "\$100"

$$SB = PB + EB$$

$$100 - x = 20$$

$$PB = 80$$

Here the private cost of 60 is not greater than private benefit of 80.

D) Since the **net social value is "0"**, **SB will be equal to SC**.

2-A:

A) Allocative efficiency occurs when resources are allocated optimally from the society's point of view. This occurs when marginal cost pricing is done that is when $MC = AR$. This can be shown through the equilibrium "e" and the corresponding price level will be "A"

B) B represents the price level which corresponds to the normal profit equilibrium that is when $AC = AR$.

C) C represents the price level which corresponds to the productively efficient level of output where $MC = ATC$.

D) D represents the price level which corresponds to the profit maximizing level of output where $MC = MR$.

3-B:

A government is considering building a new railway line in its country. It has to choose one of four high-speed routes.

The benefits and costs of each route are shown below.

Which route should be chosen?

	private benefits \$bn	external benefits \$bn	private costs \$bn	external costs \$bn	SC	Net SB
A	14	20	0.4	0.1	0.5	33.5
B	16	22	1.6	0.4	2.0	36.6
C	18	12	0.2	1	1.2	28.8
D	20	18	2	4	6	32

Handwritten notes:

- * Net social benefit $\rightarrow SB - SC$
- * project with the highest net social benefit will be selected.
- * $SB = PB + EB$
- * $SC = PC + EC$

The route which yields the highest net social benefit will be chosen.

Marginal utility is the change in total utility resulting from consuming one more unit of a good. In this case, it's the change in total utility from consuming one more apple.

Here's the marginal utility calculation:

- Between 0-1 apples: $MU = 10 - 0 = 10$
- Between 1-2 apples: $MU = 18 - 10 = 8$
- Between 2-3 apples: $MU = 24 - 18 = 6$
- Between 3-4 apples: $MU = 28 - 24 = 4$
- Between 4-5 apples: $MU = 30 - 28 = 2$
- Between 5-6 apples: $MU = 30 - 30 = 0$

Therefore, the correct answer is D) 6.

* Substitution effect: $M - N$
 * Income effect: $N - A$
 * Income effect outweighs the substitution effect

good Y
 J
 O
 A
 N
 M
 B
 C
 D
 K
 L
 I
 good X

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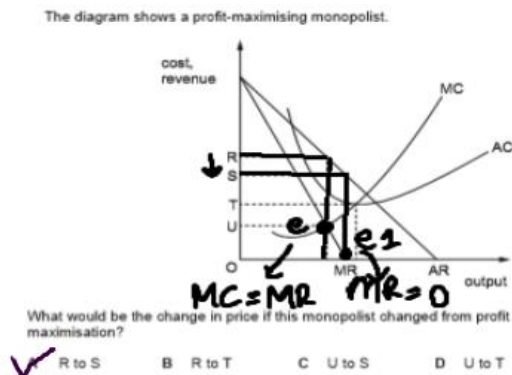
given good

D represents a "normal good"

D) Horizontal integration refers to the merger or acquisition of two firms that operate at the same stage of production or distribution, often in the same industry. In this case, two breweries merging is an example of horizontal integration, as they are both in the same industry (beer production) and are combining their operations. **This explains why key B will be correct.**

C) A cotton mill doubling its output is an example of internal growth, not integration.

7-A:



✱ At the profit-maximizing level of output ($MC=MR$), the price of "R" was charged. But, at the revenue maximizing level of output (where $MR=0$), the price of "S" is charged, this means that the price changes from "R to S"

✱ As a result of which, key A, becomes correct.

8-B:

A – The goal of **"profit-maximization"** (achieved when $MC=MR$) represents that the firm X wants to make the highest possible level of profits, which is not correct.

B – This represents **'profit-satisficing'**. A satisficing behavior is an alternative business objective to maximizing profits. It means a business is making enough profit to keep shareholders happy or it is sufficient for investors to maintain confidence in the management it appoints. This is why, key B, is correct.

C – Revenue maximization **does not** represent that the goal of the firm is **"to make a minimum , acceptable level of profit"**

D – **Sales maximization**(which represents the maximization of units that are sold), **does not** represent that the goal of the firm is **"to make a minimum , acceptable level of profit"**

9-D:

Normal profits, also known as zero economic profits, refer to the level of profit earned by a firm that is just sufficient to cover all of its costs, including both explicit costs (such as wages, rent, and raw materials) and implicit costs (such as the opportunity cost of the owner's time and capital). In other words, normal profits represent the minimum level of profit required to keep a firm operating in the long run. Normal profits are considered the baseline level of profit that a firm needs to earn to remain in business.

A: If the firm had decreasing long-run average costs, which is representative of "economies of scale", this would have resulted in higher profits rather than only "normal profits" in the long-run.

B : If the firm is a public limited company with numerous shareholders, which represents that it operates in the private sector, this also represents that the objective would have been "profit-maximization"

C: The firm being owned by a small number of financial institutions represents that the firm operates in the private sector, the objective of which is "profit-maximization". Most importantly, a monopoly is related to a single supplier.

D: This option is correct because in a highly contestable market, the threat of potential entry and competition keeps the monopoly firm's profits in check. A highly contestable market is characterized by low barriers to entry and exit, which means that potential competitors can enter the market easily and compete with the existing monopoly firm. The fear of losing market share and potential entry of new firms motivates the monopoly firm to limit its profit levels and operate at a normal profit level to avoid attracting new entrants. **An example of a contestable market is the airline industry. While there are dominant players in the market, such as major airlines, the presence of low-cost carriers and the potential for new entrants make it a contestable market. Low-cost carriers, also known as budget airlines, have entered the airline industry and disrupted the market dynamics. They offer competitive pricing, simplified services, and often target underserved or niche markets. Their presence challenges the established airlines and forces them to adjust their pricing and services to remain competitive, this is because the airline industry has relatively low barriers to entry.**

10-B:

A – Statement A is correct because firm X will not choose to produce at all given the fact that its AR is neither covering its AVC(which is essential to be covered in-order to operate in the short-run), nor ATC(which is essential to be covered in order to operate in the long-run).

B – This is incorrect because firm Y is likely to operate in the short-run and not in the long-run. This is because, while its AR is greater than AVC, it is still less than ATC. **Since this option is “not correct”, this becomes the right choice.**

C – This is correct because firm Y is likely to operate in the short-run and not in the long-run. This is because, while its AR is greater than AVC, it is still less than ATC.

D –This is correct because firm Z will operate in both, short run (since $AR > AVC$) and also, in the long-run (since $AR > ATC$)

11-B:

A: Decreasing marginal costs would have occurred when marginal product of labor would have risen.

B: All inputs changing means that “the capital inputs” are also changing, and none of them is fixed, which represents that we are operating in the long-run. Since inputs are greater than outputs, this means that the long run average costs are rising, which represents a situation of “dis-economies of scale” or “decreasing returns to scale”. This is why, this is correct.

C: Economies of scale would have occurred if output was greater than inputs.

D: All inputs changing means that “the capital inputs” are also changing, and none of them is fixed, which represents that we are operating in the long-run.

The law of diminishing returns would have occurred in the short-run, when at-least one input (capital mainly) was fixed”. This is why, this is incorrect.

12-C:

The absence of large costs associated with investment in capital assets means that new firms can enter the industry without having to incur significant sunk costs, making it easier for them to enter the market. This characteristic of an industry reduces the barriers to entry for new firms. This is why, key C, becomes correct.

A) Significant advertising expenditure by existing firms can actually create a barrier to entry for new firms, as they may not have the resources to match the advertising efforts of established firms.

B) Significant economies of scale can also create a barrier to entry, as new firms may not be able to achieve the same level of efficiency and cost savings as larger, established firms.

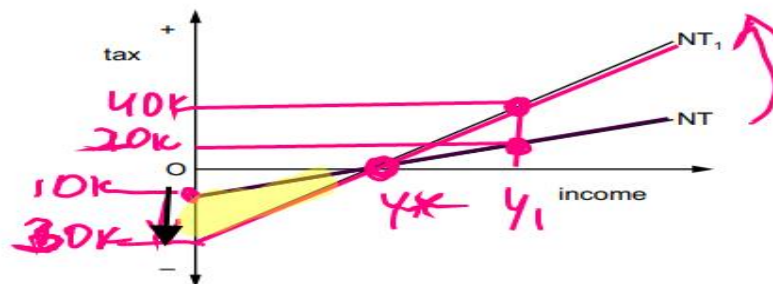
D) The possession of legal rights to produce a product (e.g. patents) can create a barrier to entry, as new firms may not be able to enter the market without infringing on the intellectual property rights of existing firms.

Option	Characteristic	Barrier Type
A	Significant advertising expenditure	Market Barrier
B	Significant economies of scale	Cost Barrier
C	Absence of large capital costs	Reduction in Cost Barrier
D	Possession of legal rights (patents)	Legal Barrier

13-D:

A country has a negative income tax.

The curve NT in the diagram shows the country's initial tax schedule.



A change in the tax rate causes the schedule to shift to NT_1 .

How will this affect work incentives and the after-tax distribution of income?

It can be seen above that the area of means tested benefits provided to those who earn less than Y^* is rising at the expense of high income groups who now have to pay a higher proportion of their incomes in taxes. This explains that the distribution of income has now become more equal.

Additionally, since the proportion of income of high income individuals taken in tax rises too, this would create dis-incentives to work, as now individuals know that a greater percentage of “additional income earned will be taken up in tax”. This explains why work incentives will weaken.

This explains why key D is correct.

14-C:



Therefore, key C, becomes correct.

Transfer earnings are the minimum income a worker needs in order to supply their labor. Economic rent is the extra income a worker receives – above the minimum level they need in order to work. → The area below the supply curve represents “transfer earnings”. The area above the supply curve and below the wage rate represents the “economic rent”

→ In a perfectly inelastic supply of labor, the quantity of a specific type of labor remains constant regardless of changes in wage rates. This is often the case for highly specialized or unique skills where there are only a limited number of individuals who possess those skills.

Unique Skills: When the supply of a particular type of labor is limited and unique, employers have little choice but to hire workers with those specific skills. This gives workers with such skills more bargaining power in negotiations, as employers need to meet their demands to retain their services.

Excess Payment: Due to the scarcity of workers with unique skills, employers offer wages that are higher than the minimum required to attract and retain them. This excess payment above what is needed to prevent them from leaving for other opportunities is considered “economic rent”

→ Therefore, the wage of such workers only comprises of “**economic rent**”

15-D:

A – There is no guarantee that ‘all’ the workers will receive the new guaranteed wage, because some will be paid higher than the minimum wage.

B – This represents an extension in quantity supplied of labor after the minimum wage legislation is enforced, but the extent to which there is an extension depends upon “price-elasticity of supply”

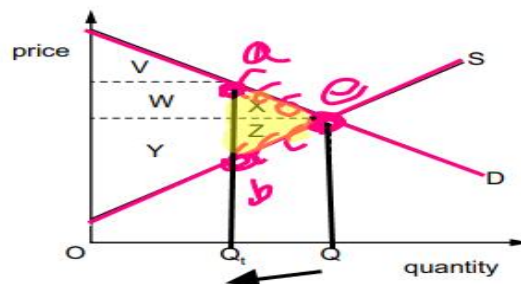
C -Although there is an extension in the supply of labor, there will be a **fall in the level of employment after the minimum wage legislation is enforced**. This is because while quantity supplied extends, there is a contraction in quantity demanded (equilibrium qty) of labor.

D – After the minimum wage legislation is enforced “above the equilibrium”, there will be a contraction in qty demanded and an extension in qty supplied of labor, representing an existence of “surplus of labor in the labor market” representing “unemployment”. Therefore, this option becomes correct.

16-C:

When the quantity is restricted from Q that is the socially optimum level of output (because $D=S$, where there is no shortage and no surplus, which represents that resources perfectly follow consumer demand) to Q_1 , there will be a welfare loss of the area $x+z$.

The diagram shows the change in a market when a government places a production quota OQ_1 on the quantity of a good that has previously traded at quantity OQ .



Which welfare loss results from the government's policy?

A $W + V$

B $W + X$

C $X + Z$

D $Y + Z$

17-B:

-> Actual growth occurs when the point inside the PPC that is the actual (denoted through “X”) moves closer to the potential that is the point on the PPC (denoted through B).

This represents effective utilization of resources, and so, key B becomes correct.

- ➔ Movement from X(actual) to A(that is a point closer to the origin), represents “recession”
- ➔ For final position to be C or D, there must have been a rightward shift of the PPC , representing “potential growth”

18-A:

GDP (Gross Domestic Product) is a measure of the total value of goods and services produced within a country's borders. It is a good indicator of the standard of living when:

1. Population is constant: GDP measures total output, so if the population is constant, it may accurately reflect the increase in an average citizen's income.
2. Inflation rate is low and stable: High inflation can distort GDP figures, making it appear larger than it actually is.
3. Income distribution is equal: If income is concentrated among a few individuals, GDP may not accurately reflect the standard of living of the majority of the population.

When these conditions are met, GDP is a more accurate measure of the standard of living in a country

19-B:

The Human Development Index (HDI) is a composite measure of three dimensions:

1. Life expectancy at birth (health)
2. Expected years of schooling and mean years of schooling (education)
3. Gross National Income (GNI) per capita (income)

A change in life expectancy directly affects the health dimension of the HDI, which is a core component of the index. This explains why key B is correct.

Here's why the other options are incorrect:

A) Average hours worked by the labor force is not a direct component of HDI.

C) Urbanization is not a direct component of HDI, although it may indirectly affect access to education and healthcare.

D) The size of the population is not a direct component of HDI, although it may indirectly affect the overall GNI per capita and thus the income component of HDI.

20-A:

A) France had the lowest rate of employment in both years.

- This statement is TRUE. France had the highest unemployment rate (11.4% in 1993 and 8.8% in 2002), which means **it had the lowest employment rate.**

B) Italy had the most employed people in 2002.

- This statement is FALSE. We can't determine the total number of employed people from the unemployment rates alone.

C) Japan more than doubled its labor productivity between 1993 and 2002.

- This statement is FALSE. There's no data about labor productivity in the table, only unemployment rates.

D) The UK created most jobs between 1993 and 2002.

- This statement is FALSE. While the UK's unemployment rate decreased significantly (from 10.5% to 5%), we can't conclude that the UK created the most jobs without additional data.

We cannot deduce any data about **number of jobs** based on **data of unemployment rates**.

21-C:

To reduce cyclical unemployment, the government can use expansionary fiscal policy, which involves increasing government spending and/or cutting taxes to boost aggregate demand and stimulate economic growth.

Option C) is the most effective policy because:

- Increasing government expenditure directly boosts aggregate demand, creating jobs and reducing unemployment.
- Decreasing income tax puts more money in people's pockets, increasing their disposable income, and encouraging consumption, which also boosts aggregate demand.

Decreasing government expenditure and increasing income tax (A) **which also represents decreasing the budget deficit (B) or increasing the government's budget surplus (D)** would reduce aggregate demand, worsening cyclical unemployment.

22-A:

The correct answer is A) changes in income inequality over time.

The Kuznets curve suggests that as an economy develops, income inequality initially increases, but eventually decreases as the economy reaches a certain level of maturity.

Here's why the other options are incorrect:

B) The Human Development Index (HDI) is a measure of well-being that combines income, education, and health. It is not directly represented by the Kuznets curve.

C) The Multidimensional Poverty Index (MPI) is a measure of poverty that considers multiple factors beyond income.

D) The Lorenz curve is a graphical representation of income distribution, and the Kuznets curve is not its inverse. The Lorenz curve plots the proportion of total income against the proportion of the population, while the Kuznets curve plots income inequality against economic growth.

23-A:

To find the multiplier, we can use the formula:

Multiplier = $1 / (MPS + MPT + MPM)$

MPS = Marginal Propensity to Save = 0.2

MPT = Marginal Propensity to Tax = 0.3

MPM = Marginal Propensity to Import = 0.3

Plugging in the values, we get:

$$\begin{aligned}\text{Multiplier} &= 1 / (0.2 + 0.3 + 0.3) \\ &= 1 / 0.8 \\ &= \mathbf{0.25}\end{aligned}$$

24-B:

The correct answer is B) high growth and low unemployment.

High economic growth and low unemployment are complementary goals because:

- Economic growth creates new job opportunities, which helps reduce unemployment.
- Low unemployment indicates a strong labor market, which can contribute to economic growth.

The other options are not necessarily complementary:

A) High growth and low inflation may not always go together, as rapid growth arising from an increase in Aggregate demand can lead to inflationary pressures.

C) A negative relationship between inflation and unemployment is outlined through the Philips curve, which represents that they are usually conflicting objectives, which **are not most likely to complement each other**.

D) Low unemployment and BOT surplus are not related. This is because when due to a fall in unemployment, people have higher incomes, they are likely to increase their consumption of imported goods. And when import expenditure rises, this results in a surplus being reduced or converted into a deficit.

25-B:

A) Liquidity preference refers to the preference of investors to hold liquid cash.

B) Quantitative easing is a monetary policy tool used by central banks to inject liquidity into the economy. By **purchasing government bonds** from the non-bank private sector, the central bank increases the money supply, reduces interest rates, and stimulates economic activity.

C) Supply-side policy refers to economic policies aimed at increasing the Aggregate supply or the productive potential of the economy.

D) The multiplier refers to the concept in macroeconomics that describes how an initial change in spending (or autonomous injection) can lead to a multiplied effect on real national income.

26-A:

Government failure refers to the phenomenon where government intervention in the economy leads to inefficient or unintended outcomes. This can occur due to various reasons, including:

- **Unintended side effects: Policies may have unforeseen consequences that outweigh their intended benefits. This may be because policies may be influenced by political considerations rather than economic efficiency.**

- **Information asymmetry: Governments may not have complete information about the market or economy.**

B) Governments can identify inefficiencies in the market, such as market failures arising from over-allocation of demerit goods.

C) Inefficient government policies do not necessarily mean that the market system is better at resource allocation.

D) Market failure does not necessarily mean that government action will improve the situation. Government intervention can sometimes exacerbate problems or create new ones, leading to government failure.

27-C:

An increase in foreign interest rates would make investments in foreign countries more attractive, leading to an outflow of capital funds from the domestic country, or **in other words, there will be hot money outflows**. This is because, investors can earn higher returns abroad, making it the “correct” option.

A) A **fall in the country's rate of inflation** represents that the returns on investment would now be anticipated. This will make domestic investments more attractive, leading to an inflow of capital.

B) An expected appreciation of the country's exchange rate would not directly lead to an outflow of capital funds. Since, appreciation results in a rise in export prices and a fall in import prices, this will contribute to a BOT deficit, which represents greater outflows in the current account. Since the capital account of the BOP remains unaffected, this option becomes incorrect.

D) An increase in foreign prices would make investment relatively safer and less risky in our country which has a relatively lower rate of inflation as inflation is likely to create uncertainty for investors. This will result in hot money inflows.

28-C:

Protecting jobs is often associated with protecting domestic industries from foreign competition. Increasing general tariffs on imports would make imported goods more expensive, reducing competition for domestic producers and potentially saving jobs in those industries which would have been lost if foreign producers outcompeted domestic producers. This explains why key C will be correct.

A) Decreasing restrictions on immigrant labor would increase competition for domestic workers, potentially leading to job losses.

B) Decreasing subsidies to domestic producers would reduce their competitiveness in the international market, as a result of which when export demand would fall, this would potentially lead to a fall in demand for labor **which is a derived demand** in those industries, as a result of which, there would be an increase in domestic unemployment.

D) Removing quotas on imports would lead to an increase in consumption of imported goods. This would lead to a fall in domestic consumption, because of which, domestic production and subsequently demand for labor would fall.

29-C:

- ➔ In an economy, a rising GDP per capita indicates that the average income of people in the country is increasing, which means people are earning more money and the economy is growing.
- ➔ A falling birth rate can also be a sign of a growing economy because it often happens when a country becomes more developed. As countries become wealthier, people tend to have fewer children. This is because they are more likely to have access to family planning services, and they may also want to focus on their careers.
- ➔ A rising household saving ratio indicates that people are saving more of their money, which can be a sign of economic growth. When people feel confident about the economy, they are more likely to save money for the future. This can also lead to more investments in the economy, which can further drive economic growth.

30-C:

Real GDP per capita is a widely used indicator of standard of living, but it has limitations. One issue is that it only accounts for formal, reported economic activity. **However, many countries have a significant informal economy (e.g., black market, unreported income), which is not reflected in official GDP figures, as a result of which GDP figures may be understated.**

Adjusting for the informal economy would provide a more accurate picture of the true standard of living in different countries. This is especially important when comparing countries with significantly different levels of informal economic activity. **This explains why key C becomes correct.**

A) Government spending is already included in GDP calculations.

B) Inflation adjustments are already made when calculating **real** GDP (“real” means adjusted for inflation)

D) Export values are also already included in GDP calculations.

